

CE 310 — Week 2 Assignment: Tails, Thresholds, and Design Choices

Due: Wed Sep 9, 9:00 AM — submit to D2L

Overview

This assignment extends the Week 2 in-class exercise into new territory. You will compute percentile thresholds that were not in the exercise, compare design implications across different exceedance levels, and write a professional interpretation of why distribution tails — not averages — drive engineering design.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Where to start	Download Wk02.06_Assignment_Answer_Sheet.xlsx from this week's D2L module. It is the template you fill in and submit — do not build your own workbook
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Where answers go	Column A of the Answers sheet labels every row — P1b_written, P2_CE_written, and so on. Each question names its row; type into the shaded cell on that row
Major declaration	Enter CE or ArcE in cell C6 — this determines which Problem 2 track is graded

Requirement	Detail
Column layout	Enter your formula in Column C , any starred (★) intermediate value in Column D , and the final answer in Column E — do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. Screenshots are not accepted
Written responses	Type them in the yellow cells lower down the same sheet — do not submit a separate PDF
Name and NetID	Type them in the Name and NetID cells at the top of the Answers sheet — that is what matches your work to your student record. A 5-point deduction applies if the NetID is blank.
File name	CE310_W2_Assignment_<NetID>.xlsx e.g. CE310_W2_Assignment_abc123.xlsx
Submission	Upload the .xlsx file to D2L → Assignments by 9:00 AM, Wed Sep 9

Problem 1 — Cold-Weather Thresholds and Heating Design (30 pts)

The Week 2 in-class exercise focused on the hot end of Tucson's temperature distribution (95th percentile cooling). This problem addresses the cold end — relevant for heating system design and freeze protection.

1a. How many hours in 2023 did Tucson's temperature fall **below** each threshold? One COUNTIF per threshold.

Row	Quantity
1a-i	Hours below 40 °F
1a-ii	Hours below 32 °F (freezing)

Put the count in Column E and the formula, typed as text, in Column C.

Your exercise did not compute any below-threshold counts — the exercise focused on the hot tail ($>95^{\circ}\text{F}$, $>100^{\circ}\text{F}$). These are new computations.

1b. The **5th percentile** of temperature is the heating design analog to the ASHRAE 1% cooling design temperature from Week 1. It represents the temperature below which only the coldest 5% of hours fall.

Calculate the 5th percentile of temperature using PERCENTILE.

A small commercial building in Tucson has a heat loss rate of $Q = U \times A \times \Delta T$, where: - $U = 0.4 \text{ BTU}/(\text{hr}\cdot\text{ft}^2\cdot^{\circ}\text{F})$ (overall thermal conductance) - $A = 5,000 \text{ ft}^2$ (building envelope area) - $\Delta T = \text{indoor setpoint } (70^{\circ}\text{F}) \text{ minus outdoor design temperature}$

Calculate the heating load (BTU/hr) for: - **Approach A (wrong):** Size the heating system for the annual average temperature (72°F). - **Approach B (correct):** Size for the 5th percentile temperature.

Note for Approach A: Tucson's annual average temperature (72°F) is higher than the indoor heating setpoint (70°F). This means $\Delta T = 70 - 72 = -2^{\circ}\text{F}$, and the formula gives a **negative heating load** — the math implies the building is losing heat to the outdoors (i.e., cooling is needed, not heating). A negative result is not a calculation error. It is the point: average-based design says no heating system is needed at all, which is demonstrably wrong on winter mornings. Report the negative value, then contrast it with Approach B.

Find the load difference. In 2–3 sentences, explain what happens to building occupants on the coldest mornings of the year if the installed system can only handle Approach A's capacity.

Enter your response in the shaded cell on row P1b_written of the Answers sheet.

1c. Your exercise computed the temperature IQR ($B3 \approx 28^{\circ}\text{F}$). Calculate the **lower outlier fence** for temperature:

Lower fence = $Q1 - 1.5 \times \text{IQR}$

How many hours in 2023 fell **below** this lower fence? Use COUNTIF with your fence value as the threshold.

Report the count you actually get. Whatever it comes to is the answer — do not assume it has to be a large number, and do not move the fence until it becomes one.

In 1–2 sentences, interpret what your count says: does the $1.5 \times \text{IQR}$ rule flag any cold hours in Tucson at all, and if the coldest hours of the year do not register as statistical outliers, does that settle whether an engineer can design as though they will not happen?

Enter your response in the shaded cell on row P1c_written of the Answers sheet.

Problem 2 — Engineering Interpretation (18 pts) — answer ONLY your track

Declare CE or ArcE in cell C6 of your Answer Sheet. That declaration determines which track below is graded — complete only its rows and leave the other track's rows blank.

Track A — Civil Engineering: Wind Design Percentile Comparison

Your exercise computed the **95th percentile** wind speed ($B10 \approx 17.2$ mph) and the wind pressure it produces ($B11 \approx 0.985$ psf), using the ASCE 7 velocity pressure relationship:

$$q = 0.00256 \cdot K \cdot V^2$$

with $K = 1.3$ (a combined exposure and gust factor) and V in mph. This track moves to a more demanding design level.

2A-a. Compute all three of the following:

Row	Quantity	Points
2A-a-i	99th percentile wind speed, using =PERCENTILE(C2:C8761,0.99)	5
2A-a-ii	Wind pressure at that 99th percentile speed, using the formula above	5
2A-b	The ratio of the 99th percentile pressure to your exercise's 95th percentile pressure (B11)	5

Note: Wind pressure grows with the *square* of speed. Before you compute the ratio in 2A-b, predict it from the speed ratio alone — then check whether your computed pressure ratio matches that prediction. The relationship between the two is the point of this part.

2A-c. By definition the 99th percentile is exceeded about 1% of the time and the 95th about 5% — roughly 87 versus 435 of the 8,700 hours with valid wind readings. In 2–3 sentences, explain what your 2A–b ratio means for a designer choosing between the two levels: how much additional structural capacity does moving from the 95th to the 99th percentile demand, and is that proportional to the difference in exceedance hours?

2A-d. Reference ASCE 7-22's use of design wind speeds tied to risk categories. In 2–3 sentences, explain why a hospital and a storage shed are not designed to the same exceedance probability, using your two pressure values as the concrete illustration.

Track A students: enter both written responses in the shaded cell on row P2_CE_written of the Answers sheet.

Track B — Architectural Engineering: Seasonal Humidity Analysis

Your exercise computed the annual humidity 90th percentile (B12) and August average humidity (B13 $\approx$ 39.95%). This track adds a winter-vs-summer seasonal breakdown and a summer design value.

2B-a. Compute all three of the following.

The two seasonal averages are the same pair you computed in **Week 1, Problem 2a**. Carry them forward from that workbook or recompute them — either is fine, and they are here as the baseline that 2B-c argues against, not as the new work. 2B-a-iii is the new value, and it carries most of the weight.

Row	Quantity	Points
2B-a-i	Winter average relative humidity — December (12), January (1), February (2)	2
2B-a-ii	Summer average relative humidity — June (6), July (7), August (8)	2
2B-a-iii	Summer 90th percentile relative humidity — the design value for 2B-c	11

For the two seasonal averages, use three separate AVERAGEIF calls, one per month, then average the three results:

$$=(\text{AVERAGEIF}(E2:E8761,12,D2:D8761) + \text{AVERAGEIF}(E2:E8761,1,D2:D8761) + \text{AVERAGEIF}(E2:E8761,2,D2:D8761)) / 3$$

Note: Averaging the three monthly means is not quite the same as pooling all winter hours, because the months differ in length. Either method is accepted; the two answers differ by less than 0.1 percentage points.

For 2B-a-iii, restrict the percentile to summer hours only. Either route below gives the same answer; both build the column of summer values beside your data, not in a corner of the Answers sheet.

Filter and paste — the Week 1 method. This is the Chart 2 procedure from the Week 1 exercise, pointed at your Month column instead of at dates. Click any cell in your data, then **Data** → **Filter**. Open the dropdown on Month, untick (**Select All**), and tick **6, 7 and 8**. You should see **2,208 rows**. Copy the visible Relative_Humidity_pct values, paste (Values only) into the first empty column beside your data, switch the filter off, and run =PERCENTILE() over what you pasted.

Or a helper column, if you would rather not filter. Head the first empty column Summer_RH, fill =IF(OR(\$E2=6,\$E2=7,\$E2=8), \$D2, "") down all 8,760 rows, and run =PERCENTILE() over the whole column. The "" it leaves on every non-summer row is text, and PERCENTILE skips text, so pointing at all 8,760 rows still reads only the 2,208 summer values. IF gets a proper introduction in Week 3 — you are welcome to use it now, but nothing this week requires it.

Either way: if your answer lands near 64% you have taken the percentile over the whole year instead. Check that only June, July and August hours reached the column.

2B-b. Your results will show that **winter average humidity is higher than summer average humidity** in Tucson — the opposite of what most students expect. In 2–3 sentences, explain this counterintuitive result. Reference the North American Monsoon and the role of Pacific moisture in Tucson winters.

2B-c. Despite winter having the higher *average*, ASHRAE engineers size cooling-season dehumidification for *summer* conditions. Compare your 2B-a-iii summer 90th percentile against your 2B-a-i winter average, and in 2–3 sentences explain why the summer design percentile — not the higher winter mean — governs equipment selection. Reference ASHRAE 62.1 and the concept of design conditions.

Track B students: enter both written responses in the shaded cell on row P2_ArcE_written of the Answers sheet.

Problem 3 — Reflection (12 pts)

Answer in **one paragraph (100–150 words)**:

A colleague argues: *“Choosing the 95th vs. 99th percentile as the design threshold is a minor technical detail that engineers debate too much — the difference is only a few percent of hours, so the practical impact on design must be small.”*

Using at least **one specific number from your analysis** (a pressure ratio, an exceedance count, or a computed threshold), write a professional response that either agrees, disagrees, or qualifies this argument. Address whether the “few percent of hours” framing is the right way to evaluate the impact of the design choice.

Enter your response in the shaded cell on row P3_written of the Answers sheet.

Grading Summary

Problem	Points
Problem 1a — 2 cold thresholds with COUNTIF formulas (5 pts, then 4 pts)	9
Problem 1b — Heating load comparison + written explanation	12
Problem 1c — Lower outlier fence + interpretation	9
Problem 2 — Engineering Interpretation, by track (Track A wind percentiles / Track B seasonal humidity) — 15 computed + 3 written, same total either way	18
Problem 3 — Reflection paragraph	12
Total	60 pts